Tariffs Abroad, Transfers at Home
How Britain Absorbs a Trade Shock*

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Abstract

How would alternative direct labour-income responses to a foreign tariff pass through the UK tax–benefit system? I construct sector stress scenarios from the 2025 tariff schedule and US-export intensity by SIC division, then apply wage, displacement, inactivity and reallocation margins to Family Resources Survey households in PolicyEngine UK. The exercise is static and first-round: neither the tariff-to-earnings mapping nor the adjustment margin is causally estimated. Its central object is the decomposition of gross earnings losses into disposable-income and fiscal responses under statutory rules. Displacement is assigned independently at the worker level, and Universal Credit claiming is re-drawn consistently for changed benefit units that are entitled after each shock. Results therefore describe conditional scenarios, with assignment dispersion reported separately from statistical uncertainty; they are not estimates of the tariffs' total macroeconomic or household incidence.

Keywords: tariffs, trade shocks, microsimulation, poverty, inequality, public finances
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1 Introduction

This paper is a static, first-round stress test of the direct labour-income channel. It does not identify the tariffs' total causal household or macroeconomic effect; the tariff-to-earnings mapping and adjustment margins are scenarios evaluated under statutory tax–benefit rules. Its contribution is a UK household application to the 2025 foreign-tariff shock, complementing existing macro–micro and tariff-incidence work.

Who bears a foreign tariff shock after the tax–benefit system responds—and how much does the answer depend on whether workers adjust through job loss, wage cuts in place, exit into economic inactivity, or reallocation into lower-paid services work? In April 2025 the United States imposed a 10 per cent baseline tariff on UK goods, with 25 per cent rates on cars and steel; the May 2025 Economic Prosperity Deal (EPD) cut the auto rate to 10 per cent on a 100,000-vehicle quota, offered conditional steel relief, and—in a December 2025 side agreement paid for through NHS pricing concessions—exempted pharmaceuticals. A rich institutional literature has traced the macroeconomic, sectoral, regional and firm-level consequences of this shock (OBR, 2025; Bank of England, 2025; City-REDI, 2025; Centre for Cities, 2025); separate tariff studies map consumer prices and income exposure into heterogeneous household welfare (Artuc et al., 2021; Luu et al., 2020). The narrower missing link is the UK partner-country labour-income channel: no analysis passes this tariff shock through the statutory tax–benefit system to disposable income, poverty and the Exchequer. This paper fills that cell.

The design is primitives-first. The shock enters as the tariff schedule itself, combined with US-export intensity by SIC division (ONS/HMRC trade-by-SIC) and an export-demand elasticity, to yield sector-level earnings shocks; the elasticity (2.0) is anchored on the realised outturn—US-bound goods exports fell 24.7 per cent in the first full tariff month against a 12.8 per cent trade-weighted tariff—and cross-checked against UK worker-level trade-shock estimates (De Lyon and Pessoa, 2021) and the OBR and Ignatenko et al. (2025) aggregate anchors. Employment is an *outcome* of the derived shock, not a free input. The derived sector shocks are then realised on Family Resources Survey workers in exposed industries under four adjustment-margin families: *displacement* into unemployment (the China-shock short run of Autor et al., 2013), *wage cuts with reallocation* into other sectors (the German and Danish long run of Dauth et al., 2021 and Traiberman, 2019), and *exit into inactivity* for older displaced workers—the margin UK deindustrialisation actually chose, when coalfield and steel job losses became benefit-financed economic inactivity rather than measured unemployment (Beatty et al., 2007; Beatty and Fothergill, 2017), and *reallocation into services*—the same displaced workers re-employed in retail, health, land transport and food service at an FRS-calibrated 28.3 per cent annual-earnings penalty, the margin the China-shock evidence actually describes (Autor et al., 2014; Dauth et al., 2021). The first three hold the aggregate earnings loss fixed; the fourth removes only the wage penalty on movers, and so is level-comparable with displacement (identical worker sets, paired draws) and rate-comparable with all. PolicyEngine UK, an open-source tax–benefit microsimulation model, computes disposable income, poverty, inequality and the Exchequer position under each.

The central 50-assignment results show substantial assignment dispersion. Under the full schedule, Bernoulli displacement affects $35{,}313 \pm 10{,}081$ weighted workers, removes £1.773 billion $\pm$ 0.652 billion of earnings, and is cushioned by 35.6 ± 4.4 per cent; the earnings-equivalent wage-cut scenario is cushioned by 41.0 per cent. The corresponding Exchequer costs are £0.804 billion $\pm$ 0.355 billion and £0.985 billion. Relative BHC poverty rises by 0.043 ± 0.023 percentage points under displacement and does not move under wage cuts. Under the EPD schedule, displacement affects $27{,}531 \pm 9{,}555$ workers, costs the Exchequer £0.635 billion $\pm$ 0.338 billion and raises poverty by 0.034 ± 0.016 points. All $\pm$ values are assignment SDs, not standard errors or confidence intervals.

Two sectoral nuances discipline the headline claims. Steel carries a caveat: roughly 80 per cent of UK steel exports go to the EU, so the steel–Wales cell is more exposed to EU/UK safeguards than to US tariffs. And pharmaceuticals—exempted under the December 2025 deal in exchange for VPAG rebate cuts and NHS pricing concessions—is a fiscal and pricing channel, not an employment channel, and is treated as such. Finally, an EPD counterfactual (full tariffs versus deal-mitigated rates) prices what the deal was worth, sector by sector and region by region, to households and the Treasury. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment signs; we compute first-round fiscal incidence holding fiscal policy fixed.

The remainder of the paper proceeds as follows. Section 2 situates the study between the quantitative trade-incidence literature and tax–benefit microsimulation. Section 3 sets out the tariff schedule, the trade-by-SIC exposure mapping, the derived sector shocks and the four adjustment-margin families. Section 4 presents the results: the derived shocks and their outturn anchor, the cushioning result and its take-up contingency, the distributional, fiscal and regional incidence by margin, and the EPD counterfactual. Section 5 turns to the EPD counterfactual and policy responses. Section 6 concludes.

2 Literature

This paper sits at the intersection of four literatures: the quantitative and reduced-form work on the incidence of trade shocks; the evidence on how workers actually adjust to them; the institutional analyses of the 2025 US tariffs on the UK; and the microsimulation literature on tax–benefit cushioning of earnings shocks. Existing macro–micro studies impose labour-market shocks on tax–benefit models, while tariff studies map price and income exposure into heterogeneous household welfare. The World Bank's Household Impacts of Tariffs framework combines household consumption and income portfolios (Artuc et al., 2021); the OECD links its METRO trade model to household-budget surveys through an explicit GTAP–COICOP concordance (Luu et al., 2020). The narrower contribution here is a UK application that maps the 2025 partner-country export-demand exposure into alternative labour-income stress scenarios and then through the statutory tax–benefit system. It is a first-round scenario exercise, not a causal estimate of the tariffs' total household incidence.

2.1 Trade shocks and their incidence

Two recent papers anchor the structural end and frame the adjustment-margin question as its poles. Ignatenko et al. (2025) build a quantitative general-equilibrium model—nesting Armington, Eaton–Kortum, Krugman and Melitz structures with exact hat algebra over 194 countries—in which the 2025 tariffs enter as a *primitive* on trade shares. Wages are flexible and labour supply elastic, so employment is an equilibrium *outcome*; strikingly, its sign flips with revenue recycling (a tariff-financed income-tax cut raises US employment by 0.32 per cent, a lump-sum rebate lowers it by 0.41 per cent), and the optimal US tariff is a uniform 19 per cent independent of bilateral deficits. The UK is in their sample, and their replication data provide the UK-specific welfare and employment cells we use as aggregate anchors. Atkin et al. (2025), by contrast, apply a Baqaee–Farhi wedge framework to Chilean administrative data with factor supplies *fixed*: there is no employment margin at all, all adjustment runs through flexible wages and prices, and household incidence arrives via consumption baskets, factor ownership, profits and transfers—their "statutory versus economic incidence" framing is the one we adopt. One paper derives employment from the tariff primitive; the other assumes it fixed and lets wages absorb everything. A third structural anchor sits between the poles: Rodríguez-Clare et al. (2025) embed the 2025 trade war in a dynamic trade-and-reallocation model with *downward nominal wage rigidity*, in which higher tariffs raise US manufacturing employment but lower total employment as falling real wages depress participation, with US real income down about 1 per cent by 2028—rigid wages turn the same primitive into an employment shock that flexible-wage models turn into a wage shock, which is precisely the disagreement our adjustment-margin axis makes an explicit scenario dimension. On the partner-country side, Michelena et al. (2025) run the 2025 escalation through a multiregional input–output framework and find employment and export losses concentrated in the *targeted* countries—the same statutory-versus-economic distinction at the country level, but at the level of sectoral aggregates rather than households. Complementing the model-based work, den Besten and Känzig (2025) construct a narrative series of US tariff shocks over 1840–2024 and find tariff increases robustly contractionary—imports fall sharply, exports decline with a lag, and output drops persistently—historical discipline for treating the shock as a negative export-demand shock rather than a benign relative-price change.

The reduced-form lineage begins with Autor et al. (2013): import competition caused large, persistent employment and wage losses in exposed US commuting zones, with effects persisting to 2019 and no offsetting migration (Autor et al., 2021). Caliendo et al. (2019) provide the structural counterpart—a dynamic spatial model with costly mobility in which the China shock is a productivity/trade-cost primitive, generating aggregate gains alongside concentrated regional losses—and Adão et al. (2023) supply the theory-consistent mapping from regional shift-share estimates to aggregate counterfactuals that cautions against reading regional results as national ones. Kim and Vogel (2021) translate reduced-form employment and wage responses into money-metric worker welfare, finding regressive incidence with displaced workers' losses exceeding wage losses; Galle et al. (2023) show in a Roy-model setting that aggregate gains coexist with group-specific losses. Our contribution relative to this structural incidence work is to replace stylised worker groups and stylised transfers with real households and the actual UK tax–benefit system.

The 2018–19 US tariff episode supplies the sharpest direct evidence on who pays a tariff. Pass-through into US import prices was essentially complete, placing the incidence on the importing country's firms and consumers (Amiti et al., 2019; Fajgelbaum et al., 2020), with retail prices adjusting more slowly than border prices (Cavallo et al., 2021)—a pattern the 2025 wave repeats: matching daily retailer microdata to product-level tariff rates, Cavallo et al. (2025) estimate retail pass-through of about 20 per cent by September 2025, contributing roughly 0.7 percentage points to the US CPI; and Flaaen and Pierce (2019) show that US manufacturing employment fell on net, as input-cost increases and foreign retaliation outweighed import protection. The retaliation side of that episode is the closest analogue to our channel—a partner's tariff arriving as an export-demand shock: Waugh (2019) finds Chinese retaliatory tariffs depressed consumption in exposed US counties, and Blanchard et al. (2019) trace the electoral fallout of retaliatory tariffs across US counties. The broader distributional literature—surveyed in Fajgelbaum and Khandelwal (2016) and Fajgelbaum and Khandelwal (2022), with Borusyak and Jaravel (2021) decomposing expenditure against earnings channels—establishes that trade shocks redistribute along both consumption and income margins. All of this work prices tariffs at the border, the store, or the county; none of it passes the shock through a household-level tax–benefit system, which is the step taken here. Two propagation channels our direct-channel design deliberately excludes have well-measured analogues in this literature. Firm-level shocks travel along supply chains—Barrot and Sauvagnat (2016) show idiosyncratic disruptions imposing substantial output losses on customers of affected suppliers, and Acemoglu et al. (2016) find network effects on aggregate outcomes several times larger than the direct effect—and tradable-sector job losses multiply into local services, with each lost tradable job costing roughly 0.4–1.6 additional local jobs across the US and German estimates (Moretti, 2010; Gathmann et al., 2020). Both channels scale our direct-channel numbers upward; Section 6 carries the caveat. Open-economy macro adds an offsetting margin: exchange rates absorb part of a tariff, but imperfectly and state-dependently—at most a fifth of the 2018–19 dollar appreciation is attributable to the tariffs (Jeanne and Son, 2024), the dollar appreciates on unilateral tariffs but depreciates under retaliation (Corsetti et al., 2025), and tariff *uncertainty* itself depreciated the dollar in 2025 through a risk-premium channel (Kalemli-Özcan et al., 2026); model-based analyses of retaliation similarly find non-trivial consumption losses for passive trade partners (Auray et al., 2020). We hold the exchange rate fixed and let the outturn-anchored elasticity absorb whatever FX offset in fact occurred. Early assessments of the 2025 wave itself (Fajgelbaum and Khandelwal, 2026) again find the incidence landing on the tariff-imposing country's consumers—complementary to our question, which is the incidence on the tariffed *partner's* workers.

2.2 Adjustment margins: how workers absorb trade shocks

The worker-level evidence says the margin is mostly earnings, not permanent unemployment. Autor et al. (2014) find US workers exposed to import competition lose earnings amid churn across employers and sectors rather than through indefinite joblessness. Traiberman (2019) shows for Denmark that occupation-specific human capital produces reallocation with persistent wage losses; because our shock is an *export-demand* shock rather than import competition, the most direct microeconometric precedents are Hummels et al. (2014), who instrument Danish firm-level export demand and trace it into worker wages by task, and Garin and Silvério (2024), who show that Portuguese exporters pass idiosyncratic export-demand shocks into the wages of incumbent workers rather than shedding them—firm-level evidence for exactly the wage-cut family of our scenario grid. Dauth et al. (2021) provide the cleanest evidence for the wage-margin family, with German workers reallocating into services and other plants at lower wages. Dix-Carneiro (2014) estimates that reallocation takes years, with adjustment costs of 1.4–2.7 times annual wages and older workers hit hardest—calibrating our transition horizons and age gradients—while Dix-Carneiro and Kovak (2017) show regional effects that *grow* over two decades on a joint wage/employment margin.

The UK's own history points to a third margin absent from the US and German studies: benefit-financed economic inactivity. Beatty et al. (2007) document that coalfield job losses became male inactivity on incapacity benefits, not measured unemployment, with incomplete recovery after twenty years; Beatty and Fothergill (2017) show that 1980s–90s industrial job destruction *still* inflates the deficit through higher benefit claims and lower tax receipts—the direct UK precedent for our fiscal-incidence question, retrospective and area-level where ours is prospective, rules-based and household-level. The margin is not merely historical: Roberts and Taylor (2022) show in 2009–18 longitudinal data that disability benefit claims still respond to local labour-market slack over and above health status, Beatty and Fothergill (2023) document the persistence of hidden unemployment among incapacity claimants across large parts of Britain, and the IFS records 4.2 million working-age people on health-related benefits by 2024, up from 3.2 million in 2019 (IFS, 2024)—the inactivity margin is live at exactly the moment the tariff shock arrives. Aragón et al. (2018) add gender-differentiated margins from mine closures, and Rud et al. (2022) decompose displacement costs into wage cuts on re-employment versus non-employment spells—directly calibrating the split between our displacement and wage-cut families. For UK-specific elasticities, De Lyon and Pessoa (2021) is the key source: UK workers in Chinese-import-competing industries lost employment years and earnings without equally gainful re-employment, with larger female losses through employment years (see also Dorn and Levell, 2024). Closest to our object, Irastorza-Fadrique et al. (2025) study UK *household* responses to trade shocks—added-worker effects, later retirement, self-employment—but model behaviour, not tax–benefit mechanics; we treat their margins as complementary to our rules-based incidence.

2.3 The 2025 tariffs and the UK

The institutional literature on the 2025 shock is rich but stops above the household. NIESR's NiGEM scenarios and the OBR's March 2025 tariff analysis provide the macro anchors—the OBR puts the permanent UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent (NIESR, 2024; OBR, 2025). The Bank of England estimates the effective US tariff on the UK at roughly 9 per cent (against about 18 per cent globally) and finds trade diversion has mildly *lowered* UK import prices (Bank of England, 2025; Dhingra, 2025)—which is what licenses our isolating the export-demand/earnings channel while holding the price and monetary channels fixed; the Bank's own staff work on the exchange-rate response to tariffs and retaliations (Corsetti et al., 2025) supplies the FX side of that held-fixed channel. The multilateral institutions frame the aggregates: the IMF's October 2025 outlook has global growth slowing to 3.2 per cent in 2025 amid the tariff shock (IMF, 2025), and the WTO's October 2025 update puts merchandise trade volume growth at 2.4 per cent in 2025 but only 0.5 per cent in 2026 as the tariffs bind (WTO, 2025). Firm surveys find most UK firms expecting small negative effects, with the US taking about 22 per cent of UK exports (CEP, 2025). The regional and sectoral geography is sharp: Coventry sends 22 per cent of its exports to the US (Centre for Cities, 2025); auto tariffs are estimated to cost £10bn of GDP over five years with the West Midlands bearing 62 per cent of it (City-REDI, 2025); SMMT data show US vehicle exports collapsing from 7,077 units in April to 4,223 in May 2025 before the EPD's implementation (SMMT, 2025). UK Steel notes that some 80 per cent of UK steel exports go to Europe—the basis of our steel caveat—and the December 2025 pharmaceutical deal (0 per cent tariffs for three years, paid for through VPAG rebate cuts from 22.5 to 14.5 per cent and NHS pricing concessions) makes pharma a fiscal/pricing shock rather than an employment shock (ABPI, 2025). The Resolution Foundation's household-facing analysis is descriptive rather than microsimulated (Resolution Foundation, 2025a); the IFS has no dedicated distributional publication on the 2025 tariffs. Across macro, sector, region and firm, the household/fiscal cell is empty.

For UK trade-shock precedents more broadly, Dhingra et al. (2017) map Brexit trade costs to income losses, and Fetzer and Wang (2024) provide the best district-level trade-shock incidence estimates (synthetic-control output losses of 5–10 points concentrated in manufacturing districts); Fetzer (2019) and Bloom et al. (2019) supply the political-economy and uncertainty channels. On household prices, Breinlich et al. (2022) use product-level import shares to identify the pass-through of the referendum depreciation, estimating a 2.9 per cent consumer-price increase and a similar real-wage loss. Levell et al. (2017) show why incidence can differ across UK households: food has much greater import content than consumption overall and household budget shares differ. These studies benchmark a consumption-price channel that the present cash-income exercise deliberately holds fixed.

The institutional modelling benchmark is also relevant to what this exercise does and does not claim. The UK Trade Modelling Review recommends a relatively simple, empirically grounded CGE core, complemented by granular partial-equilibrium modules, regional and dynamic extensions, parameter ranges and ex-post validation (Department for International Trade, 2021). The OECD operationalises the household link by mapping METRO sector prices into household expenditure and compensating variation (Luu et al., 2020); the World Bank instead offers transparent first-order accounting of consumer and income-side exposure while explicitly excluding substitution, variety and productivity responses (Artuc et al., 2021). Relative to those economy-wide or price-side tools, this paper is a deliberately partial-equilibrium fiscal-incidence module rather than a complete household-welfare evaluation.

2.4 Tax–benefit cushioning of earnings shocks

The methodological ancestors are Dolls et al. (2012), who use EUROMOD and TAXSIM to show EU tax–benefit systems absorb on average about 38 per cent of a proportional income shock and 47 per cent of an unemployment shock (against roughly 32 per cent in the US). The EU averages are not the right comparator for this paper; their *UK* cells are, and they are close to ours: the UK income-shock stabilisation coefficient is 0.352 (income tax 0.267, social insurance contributions 0.054, benefits 0.031) and the UK unemployment-shock coefficient 0.415 (tax 0.191, SIC 0.061, out-of-work benefits 0.163). Our 41.0 per cent wage-cut rate and 35.6±4.4 per cent displacement rate bracket those levels, although the latter carries substantial assignment dispersion, which is the external validation the paper is entitled to take; where we differ is in the *composition* of the unemployment-shock total, and Section 4.2 traces that difference to the out-of-work benefit component and to the composition of the displaced population. The interpretation of the tax term as an effective *marginal* rate is theirs too—and older still: Auerbach and Feenberg (2000) originate the framing of the income tax as an automatic stabiliser operating at marginal rates, and McKay and Reis (2016) formalise how progressivity and the extensive/intensive structure of shocks determine stabilisation in general equilibrium. The asymmetry between marginal and average relief is therefore not our discovery; our application shows how that asymmetry operates for the selected exposed-sector population. A second ancestor is Sutherland and Figari (2013) on EUROMOD's stress-test use—but both apply stylised macro shocks, not trade-policy-derived ones. Our direct methodological sibling is Brewer and Tasseva (2021), who pass the Covid-19 labour-market shock through UKMOD to show that the UK tax–benefit system plus the emergency support schemes held the rise in income inequality far below the rise in earnings inequality: the same architecture—an externally derived shock realised on survey households and run through the full UK benefit rules—applied to a pandemic where ours applies it to a trade-policy shock, without the emergency schemes and with the adjustment margin as the central axis. The wider macro-micro linkage literature supplies the lineage for the shock-derivation step itself: Bourguignon et al. (2008) codify the techniques for passing macro shocks to household microdata, Peichl (2016) surveys the linkage of microsimulation to structural macro models, and Navicke et al. (2013) show how EUROMOD nowcasts impose estimated labour-market transitions on survey microdata—the same device as our displacement draws, applied there to observed rather than counterfactual shocks. On the US side, exposed commuting zones saw rising disability, retirement and welfare payments that offset only a small fraction of earnings losses (Autor et al., 2013); Hyman et al. (2024) show wage insurance raises re-employment and earnings, a natural counterfactual reform for our framework. The nearest tariff-microsimulation links are on the US side: the Tax Policy Center's tariff models allocate tariff burdens to tax units through consumption baskets in a full microsimulation model (Tax Policy Center, 2025), the Budget Lab at Yale prices the 2025 schedule's fiscal and distributional effects by income decile (Budget Lab, 2025), and Acosta and Cox (2024) document that the US tariff code is built regressively, with higher rates on lower-end varieties of the same goods. Two institutional facts underpin the cushioning asymmetry our results turn on. UC's capital rules—thresholds frozen in cash terms since 2006—reduced or extinguished entitlement for around two million income-eligible families, with roughly 1.1 million standing to gain from uprating alone (Resolution Foundation, 2025b); and take-up of income-related benefits is well below 100 per cent (DWP, 2024), so statutory replacement rates overstate what displaced workers actually receive. Both bear directly on our finding that the extensive margin's cushioning is *contingent* where the in-work margin's is automatic—take-up gates entitlement inside the model and is explored only in a five-assignment sensitivity grid (Section 3.4), while the capital rules are a real-world gate the plain FRS data cannot exercise (Section 6). But all of these allocate the *consumer-price* incidence of a country's own tariffs; ours is the export-demand/earnings incidence of a *partner's* tariff. The distinction between these channels, and the absence of any analysis in ours, defines the gap this paper fills: structural incidence models have no realistic fiscal system, microsimulation has no trade shock, and the adjustment-margin axis has not been quantified by anyone. That axis is where this paper's prior gets tested. The natural reading of the two institutional facts above, and of Dolls et al. (2012)'s own unemployment-versus-income ordering, is that the system should replace a job loss more generously than an earnings-equivalent in-work wage cut, because Universal Credit exists for the former and only tapers against the latter. Section 4.2 shows that on UK microdata the two are replaced to a similar degree, and that the interesting difference is elsewhere: the two margins place different weights on automatic tax relief and claim-contingent benefits. Section 4.3 shows why.

3 Methodology

3.1 Design: primitives first

Throughout, "derived" means generated by the stated scenario mapping, not causally estimated. In particular, the April 2025 export movement used to motivate the central elasticity also reflects anticipation and front-running, prices, exchange rates, composition and other demand conditions. The $\varepsilon = 2.0$ ratio is therefore a calibration device rather than an identified export-demand elasticity. Likewise, $\pi = 1.0$ is a deliberately strong full wage-bill pass-through assumption. The parameter grid is scenario sensitivity, not a statistical confidence interval.

A reader is entitled to ask, before any number is reported, which parts of this design are read off data and which are assumed. The answer, stated in full:

Object	Status
Tariff schedule τ_j	**Data** (policy documents: the April 2025 schedule, the May 2025 EPD, the December 2025 pharmaceutical agreement).
US-export intensity x_j	**Data** (HMRC RTS/OTS customs exports over ONS ABS division turnover; Appendix A).
Export fall	**Data** in the measured family (realised HMRC outturn, May 2025–February 2026); **calibrated** in the tariff families, via an elasticity anchored on the April 2025 outturn.
Pass-through π	**Assumed** ($\pi = 1.0$), varied on a grid (Appendix D).
Adjustment margin	**Assumed**, and deliberately varied. This is the paper's free dimension.
Household incidence	**Computed** from statutory tax and benefit rules in PolicyEngine UK; no behavioural or reduced-form step.

The margin is assumed because the literature does not agree on it, and the disagreement is structural rather than empirical: Ignatenko et al. (2025) derive the employment response from elastic labour supply and flexible wages, while Rodríguez-Clare et al. (2025) derive it from downward nominal wage rigidity, so the same tariff primitive arrives as a wage shock in the first model and a job shock in the second. We do not endogenise the margin. We price the disagreement—holding the aggregate earnings loss fixed and reporting what each pole implies for households and the Exchequer. That is a weaker claim than a structural model makes, and a more auditable one.

The shock enters through primitives, not through an assumed employment effect. The pipeline is:

$$s_j \;=\; \tau_j\, x_j\, \varepsilon\, \pi, \tag{1}$$

where s_j is the derived earnings-shock size for SIC 2007 division j (the fraction of the division's employee wage bill removed by the tariff), τ_j is the tariff rate faced by the division under the schedule, x_j is its US-export intensity, ε is the export-demand elasticity, and π is the pass-through of the sector output loss to the sector wage bill. x_j is US goods exports of the division as a share of division turnover, built from the HMRC uktradeinfo RTS/OTS API (2024 US exports, £52.5 billion mapped to SIC divisions of a £55.6 billion customs-basis total, via a documented SITC-to-SIC crosswalk) and ONS Annual Business Survey turnover denominators; Appendix A gives the full provenance and validation. The elasticity $\varepsilon = 2.0$ is anchored on the realised outturn: ONS data show UK goods exports to the US falling 24.7 per cent in April 2025, the first full tariff month, against a 12.8 per cent trade-weighted average tariff implied by the schedule and the built intensities ($0.247/0.128 = 1.93$, rounded to 2.0, mid-range of short-run trade elasticities). The pass-through $\pi = 1.0$ places the full sector output loss on the sector wage bill—the appropriate short-run displacement assumption. Because ε and π enter Equation (1) only through their product, all pound-denominated results scale approximately linearly in $\varepsilon\pi$, while cushioning *rates* and margin orderings are essentially invariant to it; Appendix D verifies both properties on a 4×3 sensitivity grid over $\varepsilon \in \{1.0, 1.5, 2.0, 3.0\}$ and $\pi \in \{0.5, 0.75, 1.0\}$. Since the central $\varepsilon = 2.0$ is anchored on a single month's realised outturn (April 2025), the grid doubles as the uncertainty band around that anchor. The implied aggregate gross earnings loss under the full schedule is £1.77 billion a year, about 0.06 per cent of GDP on the direct exposed-sector channel, sensibly below the OBR's all-channel scenario range of one-third to three-quarters of a per cent (rising to about 1 per cent at peak in its reciprocal-tariff scenario) (OBR, 2025) and consistent with the UK cell of Ignatenko et al. (2025) and UK worker-level responses (De Lyon and Pessoa, 2021; Rud et al., 2022). Employment is then an *outcome* of s_j under an assumed adjustment margin—never a free input. The design covers the direct exposed-sector channel only: local-multiplier estimates of 0.4–1.6 additional local jobs per lost tradable job (Moretti, 2010; Gathmann et al., 2020) illustrate a potentially amplifying channel, while exchange rates, market switching and firm margins may attenuate the direct effect, so the derived shock is not a formal bound (Section 6). Appendix F presents an accounting sensitivity for the first omitted channel, propagating the export-demand falls upstream through the ONS domestic input-output tables via a Leontief round (Barrot and Sauvagnat, 2016; Acemoglu et al., 2016).

3.2 The tariff schedule and the EPD counterfactual

Two tariff scenarios bracket the policy. *Full tariffs*: the April 2025 schedule—10 per cent baseline on UK goods, 25 per cent on motor vehicles (SIC 29) and steel (SIC 24). *EPD*: the deal-mitigated schedule—autos at the 10 per cent in-quota rate on up to 100,000 vehicles; steel with conditional, partial relief; pharmaceuticals (SIC 21) exempt under the December 2025 agreement. The automotive scenario applies the 10 per cent rate to the modelled 2024 exposure because the quota is approximately equal to that year's US-bound volume; it does not model quota allocation, above-quota growth or an endogenous volume response around the ceiling. The difference between the two scenarios, run through identical downstream machinery, prices the EPD for households and the Exchequer. Two caveats travel with the schedule. Steel: about 80 per cent of UK steel exports go to the EU, so the US tariff is a minority driver of the sector's exposure and the steel cell should be read alongside EU/UK safeguards. Pharmaceuticals: the exemption was purchased through VPAG rebate cuts and NHS pricing concessions, so pharma's incidence is fiscal and price-side, not an employment shock; we model its employment shock as zero under the EPD and discuss the fiscal channel separately (Section 5).

3.3 Adjustment-margin families

The derived s_j is realised on Family Resources Survey 2024–25 employees in exposed divisions under four families. Wage cuts remove $\sum_j s_j B_j$ deterministically; Bernoulli displacement and inactivity remove that amount only in expectation, with realised weighted earnings varying across assignments:

Displacement (ADH short run). Within each exposed division every employee is selected independently with probability s_j, irrespective of the record's survey weight. This Bernoulli design gives each represented worker the intended probability and the sector target in expectation without forcing a weighted head-count quota. Selected workers move to unemployment with earnings, hours, pension contributions and statutory pay zeroed.

Wage cuts with reallocation (Dauth/Traiberman long run). No job loss; every employee in division j takes a proportional cut of s_j, so the weighted aggregate earnings removed equals $\sum_j s_j B_j$ by construction—exact earnings-equivalence with the displacement family's expected removal.

Inactivity (Beatty–Fothergill UK history). The displacement draw, but displaced workers aged 50 and over exit to economic inactivity, the benefit-financed route of UK deindustrialisation, rather than unemployment. The route is benefit-financed in the model, not just relabelled: every inactive worker is flagged as having limited capability for work-related activity, so their household's Universal Credit includes the LCWRA health element (£217.26 a month in 2026). In policyengine-uk 2.89.2 the survey-imputed `uc_LCWRA_element` is stored, so the post-shock benefit-unit value is set to the maximum of the stored amount and one annual statutory element; an existing recipient is never paid a second element, unaffected units are unchanged, and the run hard-errors if the element fails to reach a newly flagged person's benefit unit. This mirrors the Beatty et al. (2007) route by which industrial job loss became incapacity-benefit inactivity. The strong assumption should be stated plainly: *all* displaced workers aged 50 and over are assumed to pass the Work Capability Assessment, so the family is an upper bound on the health-element route; a sensitivity variant in which only 50 per cent of the older displaced qualify is reported alongside (Section 4.4). Work-search conditionality carries no separate fiscal machinery in the model, so the LCWRA element is the entire modelled wedge between inactivity and unemployment.

Reallocation into services (ADH/Autor et al., 2014). The same displacement draw on the same seed—so the two families are paired worker-for-worker—but the drawn workers are re-employed rather than left out of work. Their `sic_industry_division` is reassigned to one of four services destinations (SIC 47 retail, 86 human health, 49 land transport, 56 food and beverage service) in proportion to those divisions' FRS employee-weight shares (29.5, 45.8, 9.2 and 15.5 per cent), and their annual earnings are cut by a penalty calibrated on the FRS itself: weighted mean annual employee earnings of £48,272 over 2,047 hours in the exposed goods divisions against £34,594 over 1,701 hours in the four destinations, a **28.3 per cent annual-earnings penalty** that embeds the hours fall. Two variants are reported: an hours- and age-controlled **low-penalty** case of 14.0 per cent (the pure hourly-wage gap), and a **three-month lag** in which workers are out of work for a quarter of the year before the services job starts, scaling annual earnings and hours by a further $1 - 3/12$. ASHE 2024 full-time medians bracket the hours-controlled figure at 10–20 per cent; the FRS all-employee penalty is larger because it also captures the shift into part-time work. Because this family removes only the penalty on the movers rather than the full earnings-equivalent aggregate, its pound figures are comparable with the displacement family (identical worker sets, paired draws) but *not* with the wage-cut family; its cushioning *rate* is comparable with both. Section 4.9 reports the results and Appendix B the mechanics.

Two scope limits on the family set should be stated here rather than in a footnote. The wage-cut family models the *earnings consequence* of reallocation without moving anyone between sectors: workers keep their observed industry and hours, and the proportional cut stands in for the wage penalty that a completed reallocation would carry; the reallocation family does move them, but who reallocates is assumed (it is the displacement draw) rather than estimated. And no household-side behavioural margin is active—added-worker responses, earlier or later retirement, and shifts into self-employment (Irastorza-Fadrique et al., 2025) are all suppressed, as are any labour-supply responses to the tax and benefit changes themselves. All four families are therefore first-round, rules-based incidence.

The margin is the paper's central axis because the tax–benefit system treats the two poles asymmetrically: an in-work earnings reduction is offset automatically at marginal tax and taper rates, while a job loss moves the worker to the gated, means-tested extensive margin of the benefit system. How much each pole is cushioned, and—the question that turns out to discriminate between them—how much that cushioning depends on anyone doing anything, are empirical questions the microdata answer (Section 4.2).

3.4 Universal Credit take-up after the shock

The take-up experiment is applied symmetrically across margins. After each shock, changed benefit units whose all-claim UC award moves from zero at baseline to positive post-shock receive a claiming draw at the scenario take-up rate. This includes wage-cut households that cross into entitlement. Existing claimants, existing eligible non-claimants, unchanged units and post-shock-ineligible units retain their baseline flag, avoiding an unintended population-wide take-up reform. The grid therefore measures sensitivity to a common new-entitlement convention; it does not estimate actual claiming behaviour.

One modelling choice deserves its own subsection, because an earlier version of this paper got it wrong and the correction changes a headline number. PolicyEngine UK carries benefit take-up as a benefit-unit-level *stored input*, `would_claim_uc`, drawn once at dataset build time. Two properties of that draw matter. First, its target is a flat population-wide parameter—0.55 as a share of *all* benefit units, with reported UC recipients anchored to TRUE and the remainder filled to hit the target. It is not a take-up rate among the entitled, and it is therefore not comparable to the roughly 80 per cent take-up-among-entitled that the Resolution Foundation and the IFS assume; in the dataset used here the realised flag rate is 0.549 over all benefit units and 0.541 among UC-eligible ones. Second, the flag is drawn on *pre-shock* circumstances. Nothing in a naive shock pipeline re-draws it, so a worker displaced by the shock carries into unemployment the claiming decision attributed to them while they were employed and, in the great majority of cases, not entitled to anything—measured take-up among the displaced post-shock was 46.9 per cent, below even the population flag rate. The baseline flag is thus not informative about the post-shock claiming behaviour of a newly entitled family: it encodes a decision taken in a state of the world the shock has destroyed.

The pipeline therefore re-draws `would_claim_uc` after each shock only for benefit units whose circumstances changed and whose all-claim UC award moves from zero at baseline to positive post-shock, at a scenario parameter `uc_takeup` with default 0.80. The re-draw uses an independent random stream, leaving worker assignment unchanged and preserving paired comparisons. This rule applies to displacement, inactivity, reallocation and wage cuts alike; existing eligible, post-shock-ineligible and unchanged units retain their baseline flag. Take-up sensitivity is therefore evaluated symmetrically for newly entitled units, although the parameter remains assumed rather than observed.

3.5 Microsimulation and outcomes

The Monte Carlo draws describe sensitivity to the artificial assignment of displacement across weighted FRS records. Their standard deviations are assignment dispersion, not sampling standard errors, confidence intervals, or uncertainty about the tariff response, survey design or model parameters. Reported comparisons are descriptive scenario contrasts.

The realisation step—an externally derived shock imposed as status and earnings transitions on survey households, then run through the full benefit rules—follows Brewer and Tasseva (2021), who apply it to the Covid-19 labour-market shock in UKMOD, and the EUROMOD nowcasting tradition of imposed labour-market transitions (Navicke et al., 2013). PolicyEngine UK computes baseline and shocked household disposable income (HBAI cash concept throughout), relative before-housing-costs poverty, absolute before-housing-costs poverty against a fixed baseline line (computed manually, since the installed model version exposes no absolute-poverty variable; Appendix C), after-housing-costs poverty, the Gini coefficient of equivalised disposable income, decile and region breakdowns, and the Exchequer effect (foregone tax plus additional benefit spending, via the government balance). Displacement draws are Monte Carlo: every displacement/inactivity result is reported as the mean and standard deviation over n draws (default 20). Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment responses; we compute first-round incidence holding fiscal policy fixed and treat recycling as a policy counterfactual. One scope limitation on the fiscal aggregate should be stated: only direct taxes and benefits respond to the shock. Indirect taxes (VAT and duties) are not shocked, so the consumption-tax revenue foregone on the households' reduced spending is absent from the government balance and the reported Exchequer cost is an understatement of the true first-round fiscal effect.

4 Results

All central results use FRS 2024–25, period 2026, and 50 assignments. Values after ± are standard deviations across artificial assignments, not standard errors or confidence intervals. Five-assignment diagnostics are labelled exploratory.

4.1 Derived sector shocks

The full-schedule direct earnings shock is £1.77 billion annually; the EPD schedule is £1.39 billion. These are scenario inputs generated by Equation (1), not estimated causal effects. Autos have the largest full-schedule shock (5.5 per cent of the division wage bill), followed by machinery and electronics (3.6 per cent each), other transport (3.1), pharmaceuticals (2.6), and steel and electrical equipment (2.5 each).

4.2 Cushioning

Table 1: Central direct-channel scenarios, 50 assignments

Schedule	Margin	Workers (thousands)	Gross loss (£bn)	Exchequer (£bn)	Cushioning (%)
Full	Displacement	35.3 ± 10.1	1.773 ± 0.652	0.804 ± 0.355	35.6 ± 4.4
Full	Wage cut	—	1.771	0.985	41.0
Full	Inactivity	35.3 ± 10.1	1.773 ± 0.652	0.825 ± 0.355	36.9 ± 4.2
EPD	Displacement	27.5 ± 9.6	1.396 ± 0.628	0.635 ± 0.338	35.5 ± 4.7
EPD	Wage cut	—	1.386	0.769	40.8
EPD	Inactivity	27.5 ± 9.6	1.396 ± 0.628	0.651 ± 0.340	36.8 ± 4.5

Notes: Worker, gross-loss and draw-based-margin values are means ± assignment SD. Wage-cut dispersion from the common UC claiming redraw is below the displayed precision. Earnings equivalence holds in expectation, not assignment by assignment.

The full-schedule displacement and wage-cut cushioning rates differ by about five points in their means, but the displacement assignment SD is 4.4 points. The decomposition remains economically informative: proportional cuts remove the top slice of earnings and therefore receive relief near marginal tax rates; displacement removes the whole earnings stream and receives relief nearer average rates, with UC providing part of the remainder. These are descriptive model mechanisms rather than estimates of behavioural insurance reliability.

Take-up sensitivity. The regenerated grid applies the same post-shock-entitlement claiming rule to all margins. It uses only five assignments per cell and is exploratory. It shows that displacement responds more visibly to the assumed claiming probability, while wage-cut results move little because relatively few wage-cut households change UC entitlement. No grid difference is treated as statistically identified.

4.3 Mechanism

The regenerated decomposition and gate audit are also five-assignment diagnostics. Income-tax relief is larger on the wage-cut margin than on displacement; UC is more important after displacement. The plain FRS input contains no usable household-capital measure, so the model cannot establish the empirical importance of UC's capital test. New-style JSA is not activated by the imposed transition. These omissions make the gate audit a description of the implemented model, not the full UK out-of-work system.

4.4 Distributional incidence

Under full displacement, relative BHC poverty rises by 0.043 ± 0.023 percentage points and the Gini by 0.000192 ± 0.000167. Under the wage-cut scenario, relative BHC poverty is unchanged and the Gini falls by 0.000106. Inactivity raises poverty by 0.040 ± 0.024 points. The small national changes coexist with large losses for selected households, but the magnitude is assignment-sensitive.

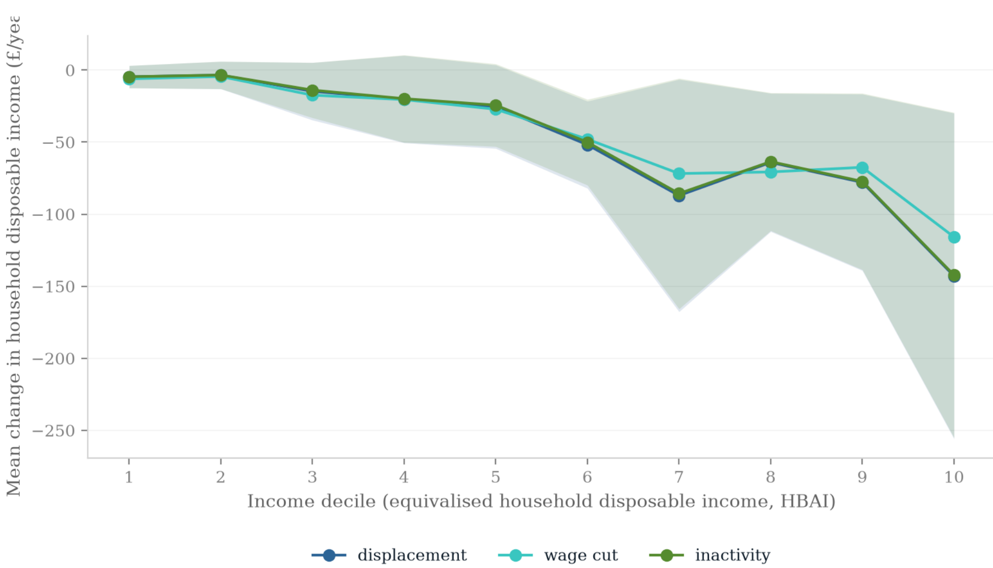

Figure 1: Disposable-income change by equivalised-income decile under the regenerated full-schedule scenarios. Shading shows assignment dispersion, not sampling uncertainty.

4.5 Who bears it

The gender, age and household-type breakdown uses five Bernoulli assignments and is exploratory. It confirms that selected exposed-sector workers are disproportionately men and concentrated in prime and older working ages, but individual subgroup estimates are too assignment-sensitive for precise rankings. The regenerated demographic artifact is reported as a diagnostic rather than as an inferential table.

4.6 Fiscal incidence

Table 2: Annual Exchequer cost (£billion)

Schedule	Displacement	Wage cut	Inactivity
Full	0.804 ± 0.355	0.985	0.825 ± 0.355
EPD	0.635 ± 0.338	0.769	0.651 ± 0.340

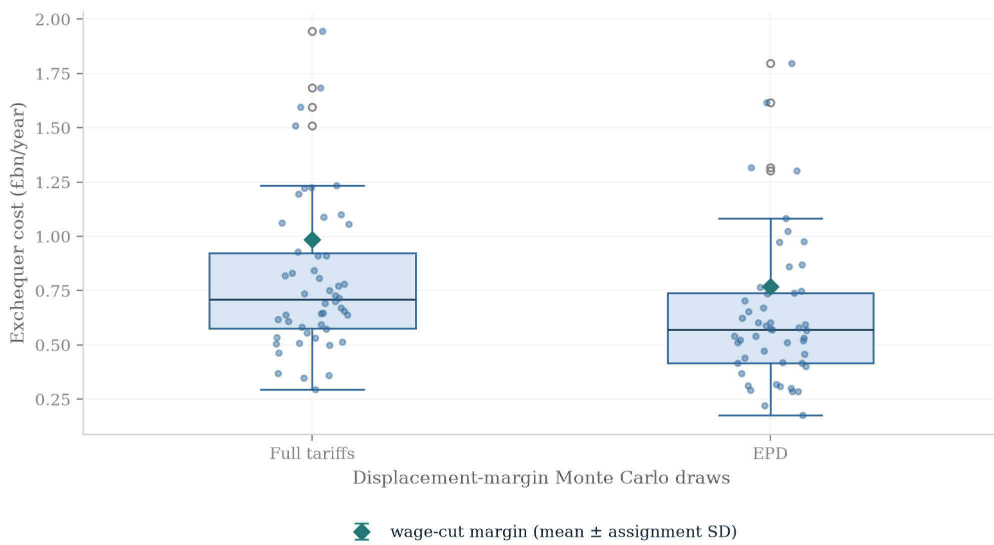

Figure 2: Annual Exchequer cost across regenerated assignments. The spread is assignment dispersion.

4.7 EPD counterfactual

On paired seeds, full minus EPD displacement differs by $7{,}782 \pm 3{,}921$ workers, £0.377 ± 0.263 billion of gross earnings, and £0.170 ± 0.144 billion of Exchequer cost. The poverty-rate difference is 0.0096 ± 0.0113 percentage points. These are paired scenario contrasts, not causal deal estimates.

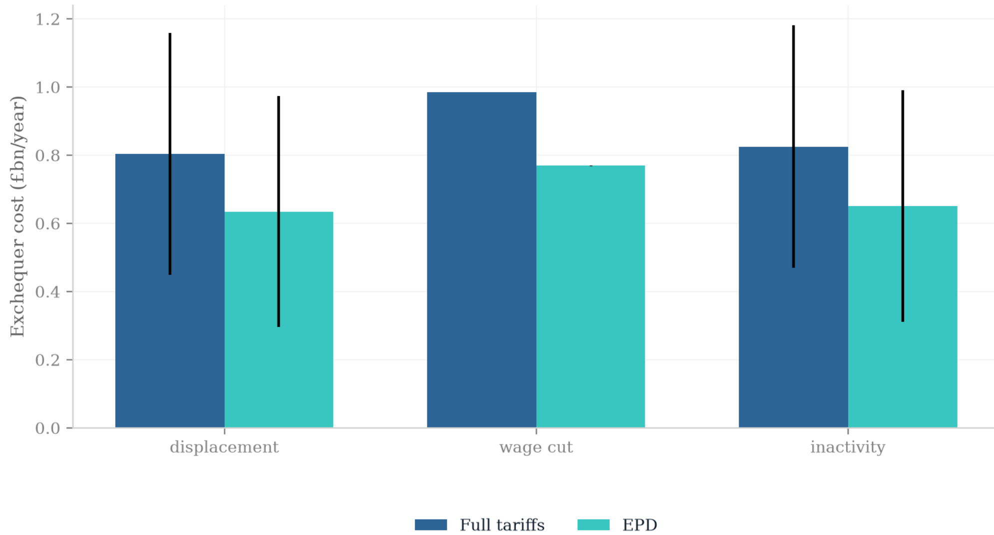

Figure 3: Regenerated full-schedule and EPD fiscal scenarios. Error bars are assignment SDs.

4.8 Regional incidence

Regional means are calculated across the same 50 Bernoulli assignments. Their dispersion is large and rankings are illustrative. They should not be interpreted as estimates of local causal exposure; BRES-linked local industry structure would be required for that claim.

4.9 Reallocation

The reallocation exercise uses five paired assignments and is exploratory. Under the full schedule, the central 28.3 per cent penalty produces a mean gross loss of £0.597 billion, Exchequer cost of £0.339 $\pm$ 0.209 billion, cushioning of 42.3 $\pm$ 3.6 per cent, and essentially zero BHC-poverty change. With a three-month lag, gross loss rises to £0.975 billion and the Exchequer cost to £0.549 $\pm$ 0.339 billion. The 14 per cent penalty gives gross loss £0.295 billion and Exchequer cost £0.171 $\pm$ 0.108 billion. Because only selected movers' penalty is removed, these pound levels are not commensurable with the economy-wide wage-cut scenario.

4.10 Validation against the outturn

The descriptive May 2025–February 2026 export window falls 12.7 per cent against its two-year same-month baseline. This is not a causal tariff estimate: it includes prices, exchange rates, front-running, rerouting and other demand conditions.

4.11 Measured outturn-shaped scenario

The measured displacement scenario affects 17,538 weighted workers on average, is cushioned by 36.0 $\pm$ 5.7 per cent, costs the Exchequer £0.429 $\pm$ 0.258 billion, raises relative BHC poverty by 0.021 $\pm$ 0.014 percentage points and raises the Gini by 0.000079 $\pm$ 0.000102. The measured wage-cut scenario is cushioned by 41.6 per cent, costs £0.528 billion, leaves relative BHC poverty unchanged and lowers the Gini by 0.000070. This is an outturn-shaped sensitivity case, not a preferred causal estimate.

5 Policy

5.1 Economic Prosperity Deal

Within the maintained direct-channel scenarios, the paired full-minus-EPD displacement contrast is $7{,}782 \pm 3{,}921$ workers, £0.377 ± 0.263 billion of gross earnings and £0.170 ± 0.144 billion of annual Exchequer cost. These SDs describe assignment dispersion and the contrasts are not causal estimates of the deal. Autos drive much of the difference; the scenario applies the in-quota rate to fixed exposure without modelling quota allocation or future above-quota trade.

5.2 Pharmaceutical channel

The EPD scenario sets the pharmaceutical employment shock to zero, but this does not make the exemption free. Its VPAG and NHS pricing concessions sit outside the labour-income microsimulation. A complete policy appraisal would cost those health-budget and price effects alongside the direct tax–benefit scenario.

5.3 Cushioning policy

The simulations support a limited conclusion: extensive-margin protection depends more on means-tested eligibility and claiming than the income-tax response to an intensive earnings loss. The five-assignment take-up grid is exploratory and does not quantify claimant behaviour precisely. Automatic enrolment after redundancy, contribution-based unemployment insurance and wage insurance are therefore relevant counterfactuals, but their costs require dedicated simulations and financing assumptions.

6 Discussion and conclusion

This paper is a conditional stress test of direct labour-income scenarios, not an estimate of the tariffs' total causal incidence. Under the full schedule, displacement is cushioned by 35.6 ± 4.4 per cent, inactivity by 36.9 ± 4.2 per cent and wage cuts by 41.0 per cent. Exchequer costs are £0.804 ± 0.355 billion, £0.825 ± 0.355 billion and £0.985 billion respectively. Every $\pm$ value is assignment dispersion.

The component mechanism is consistent with automatic-stabiliser theory: proportional earnings cuts receive tax relief near marginal rates, while a whole-job loss receives relief nearer average rates and relies more on UC. The corrected common claiming rule applies to every margin. Five-assignment take-up and gate exercises are exploratory and do not establish behavioural reliability or precise gate shares.

National poverty effects are small in the central scenarios: $+0.043 \pm 0.023$ percentage points under full displacement and zero under wage cuts. That does not make selected households' losses small. It means the exposed population is limited and often above the poverty line before the shock.

The paired full-minus-EPD displacement contrast is 7,782 ± 3,921 workers and £0.170 ± 0.144 billion of annual Exchequer cost. This is a scenario comparison driven substantially by autos, not a causal evaluation of the deal.

The main limitations are structural. The tariff-to-earnings mapping is calibrated rather than estimated; assignment is independent within broad SIC divisions; the inactivity scenario assumes WCA passage for selected older workers; New Style JSA and within-year timing are absent; household labour-supply responses are suppressed; and indirect taxes, exchange rates, market switching, retaliation and general equilibrium are outside the central model. The supply-chain calculation is an accounting sensitivity, not a lower bound, and cannot be mechanically combined with local-employment multipliers. Regional and constituency results are illustrative because local industry is not observed at the required level.

The defensible conclusion is therefore modest. For a given direct earnings loss, the adjustment margin materially changes the mix of tax relief, benefit support, poverty and fiscal cost. Quantifying the realised tariff effect requires causal trade and linked employer–worker evidence beyond this microsimulation.

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A Data appendix

US exports use HMRC uktradeinfo OTS/RTS data for 2024, mapped from SITC to SIC 2007 divisions and divided by the latest non-suppressed ONS Annual Business Survey turnover. Of a £55.6 billion customs-basis total, £52.5 billion maps to divisions after documented exclusions, chiefly non-monetary gold and precious metals. The measured window uses May 2025–February 2026 against the mean of the same calendar months one and two years earlier. Its 12.7 per cent aggregate fall is descriptive, not causal.

B Shock mechanics

Each employee in exposed division j is independently selected with probability s_j. Weighted displaced counts and earnings therefore meet targets in expectation rather than exactly. Selected workers have employment income, hours, pension contributions and statutory pay zeroed. Wage cuts remove s_j of each exposed worker's earnings deterministically. Reallocation uses the same selected workers, assigns a services destination, and applies the calibrated earnings penalty.

UC claiming is redrawn after every margin for benefit units whose circumstances changed and that are entitled post-shock. The LCWRA override sets the benefit-unit element to the maximum of its stored baseline value and one statutory annual element, preventing duplicate payment.

C Monte Carlo and measures

Central direct and measured scenarios use 50 assignments. Every $\pm$ value is an assignment SD, not a standard error or confidence interval. The survey-design, tariff-response and model-parameter uncertainty are not estimated. Five-assignment take-up, mechanism, demographic, poverty-gap, reallocation, duration, sensitivity-grid and supply-chain exercises are exploratory.

Absolute BHC poverty uses 60 per cent of the baseline equivalised median as a fixed line. The cushioning rate is $1 - \Delta Y/\Delta E$, where ΔE is gross employment-earnings loss and ΔY is household disposable-income loss.

D Elasticity and pass-through grid

The regenerated grid spans $\varepsilon \in \{1, 1.5, 2, 3\}$ and $\pi \in \{0.5, 0.75, 1\}$. Because only five displacement assignments are used per cell, it is an exploratory scaling diagnostic. Equal products $\varepsilon\pi$ generate equal deterministic wage-cut shocks; Bernoulli displacement outcomes vary across assignments. The grid is not an uncertainty interval.

E Duration sensitivity

The duration exercise uses five assignments. Its six-month construction retains half-year earnings while coding a full-period unemployed status with zero hours. This is internally inconsistent in an annual model and cannot reproduce within-year tax, waiting-period or eligibility timing. It is retained only as a rough scaling diagnostic.

F Supply-chain sensitivity

The input–output scenario adds upstream requirements using the ONS domestic Leontief inverse. The accounting shock has a 1.95 direct-plus-upstream factor. It is not a lower bound and is not multiplied by local-employment multipliers. In the exploratory five-assignment microsimulation, displacement affects 69,255 weighted workers on average, costs the Exchequer £1.801 $\pm$ 0.544 billion and raises relative BHC poverty by 0.073 ± 0.011 percentage points. The wage-cut variant costs £1.904 billion and raises poverty by 0.031 points. These figures are accounting sensitivities, not all-channel causal estimates.

G Regional incidence

H Constituency illustration

The enhanced-FRS constituency layer uses imputed industries rather than observed local industry structure. On the regenerated CSV's `income_change_gbp_per_person_full` field, the ten largest simulated losses are Ealing Central and Acton (£770), Henley and Thame (£764), Godalming and Ash (£598), Sevenoaks (£579), Orpington (£540), Chesham and Amersham (£535), Mid Buckinghamshire (£525), Hitchin (£513), Twickenham (£490) and North East Hampshire (£483) per person annually. The EPD field gives the same ordering with slightly smaller losses. These are imputation-driven illustrations, not constituency exposure estimates.

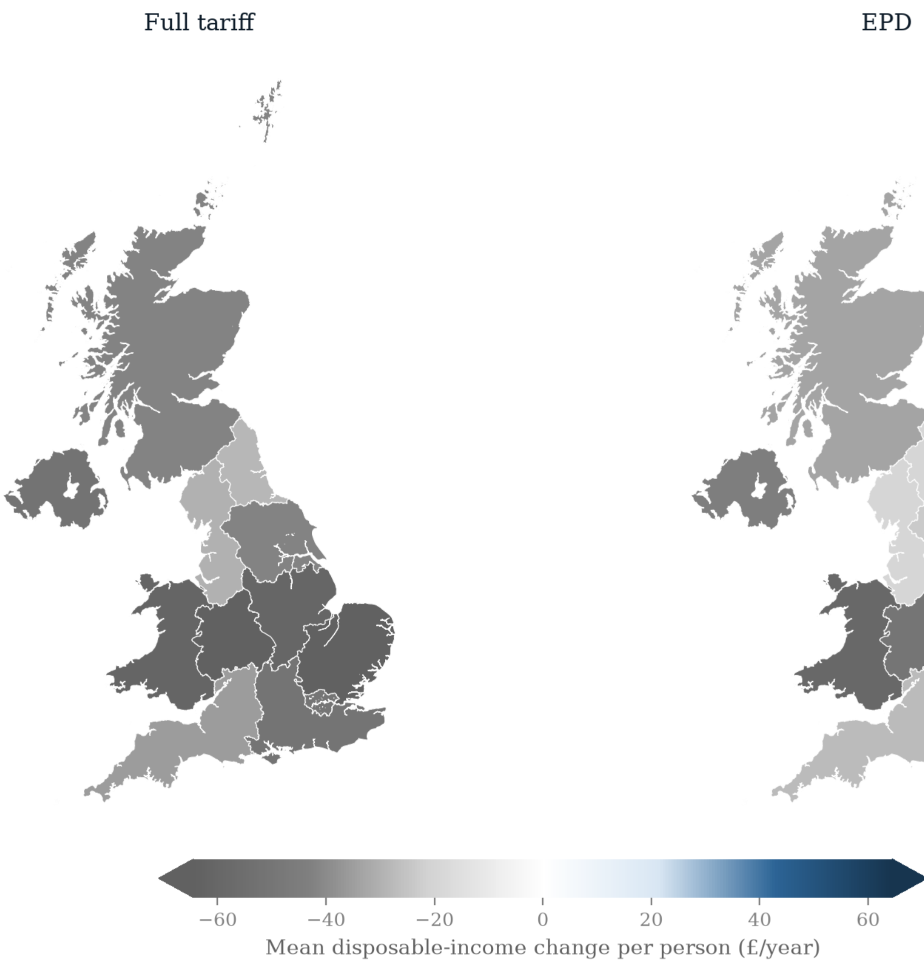

Figure 4: Regenerated regional mean disposable-income changes under full and EPD displacement. Rankings are illustrative and the underlying spread is assignment dispersion.

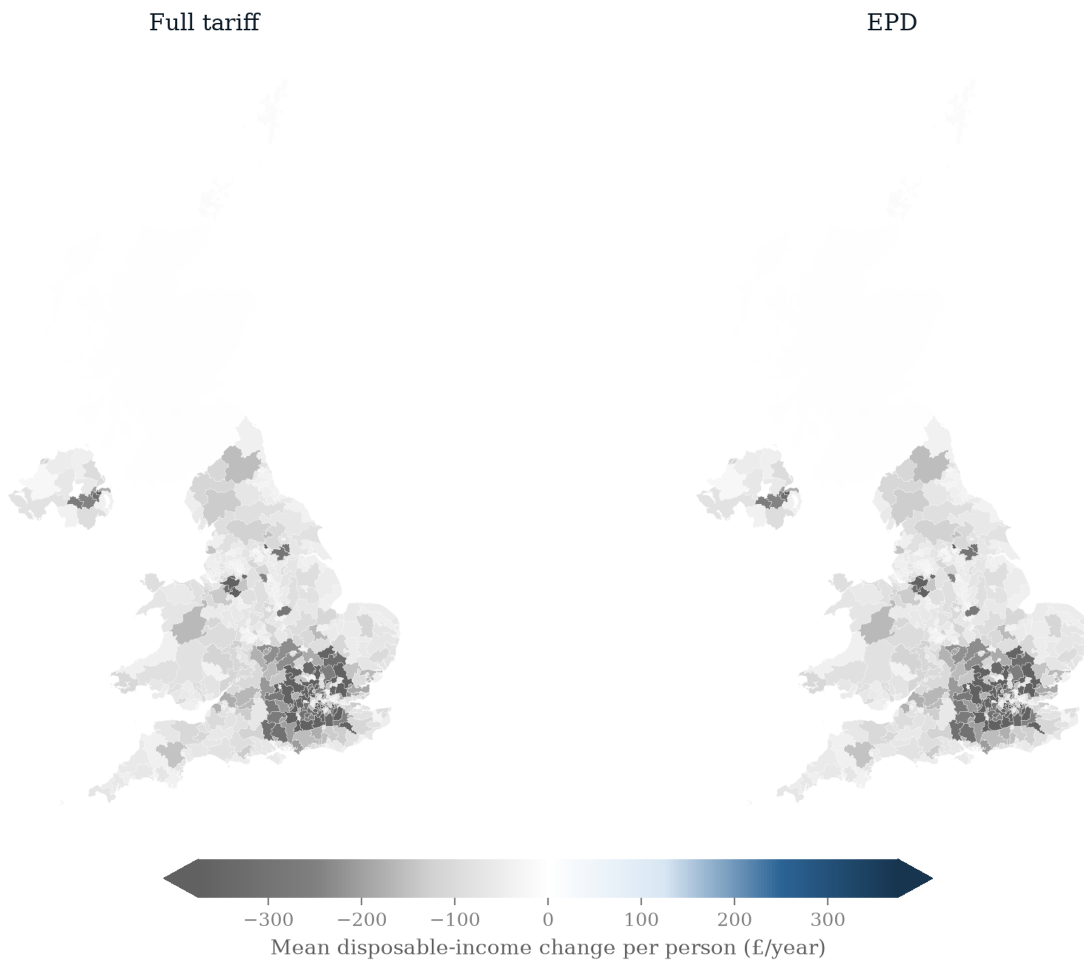

Figure 5: Regenerated constituency illustration using the current per-person income-change fields. The map does not use observed constituency industry mix.